

Page 1 - Solar Readiness

SolarSmart helps homeowners decide whether rooftop solar is a practical upgrade. The first step is not buying panels. The first step is reading the last twelve months of electricity bills and finding the home average monthly use in kilowatt-hours.

Solar works best when the roof receives direct sunlight for most of the day. South-facing roof sections usually produce the most energy in the northern hemisphere, but east and west sections can still be useful. Shade from trees, nearby buildings, chimneys, and roof vents can reduce production.

A good solar plan compares current electricity use, expected solar production, and the price paid for grid electricity. If the home uses most electricity in the evening, the owner may need a battery or a utility plan that gives credit for daytime exports.

Page 2 - System Design

A basic home solar system has panels, an inverter, mounting hardware, monitoring software, and a connection to the home's electrical panel. Solar panels produce direct current electricity. The inverter changes it into alternating current electricity that home appliances can use.

The monitoring app helps owners see daily production, peak production hours, and possible faults. If production suddenly drops on a clear day, the system may have shade, dirt, equipment trouble, or a connection issue.

Batteries are optional. A battery stores extra solar energy for evening use or backup power. However, batteries increase cost, so they are most useful when backup power is important or when the utility gives low credit for exported solar energy.

Page 3 - Proposal Details

Review the proposal details below before comparing warranty, production, and payback.

Installer Proposal Comparison

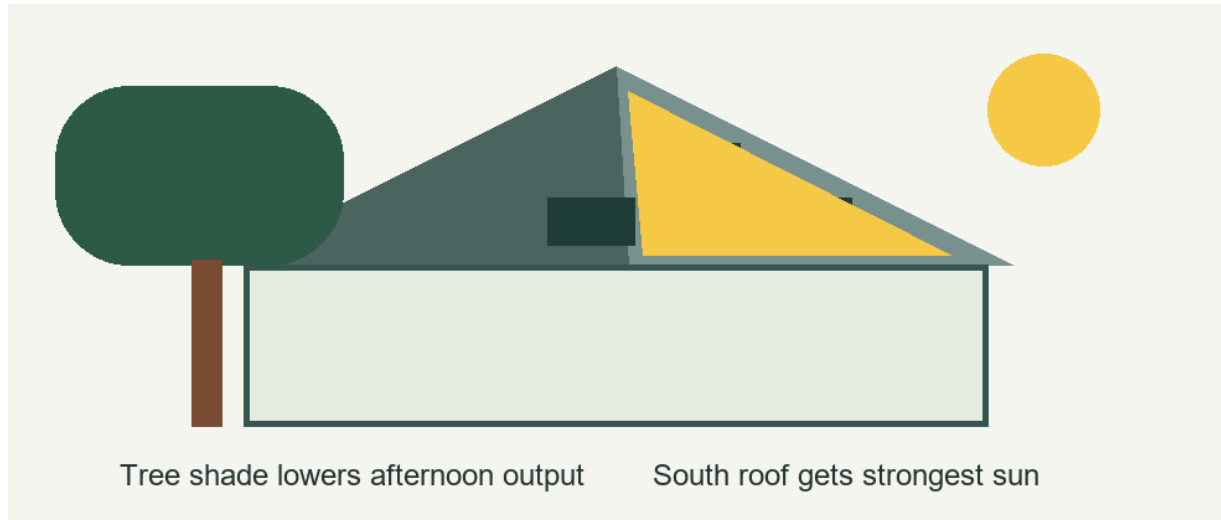
Proposal	System size	Warranty	Production	Payback
A	6.4 kW	25 yr panels	8,900 kWh	8.5 yr
B	7.1 kW	30 yr panels	9,600 kWh	9.2 yr
C	5.8 kW	25 yr panels	7,900 kWh	7.8 yr

Monthly Usage Snapshot

Month	Grid use	Solar estimate	Bill impact
April	620 kWh	540 kWh	Medium saving
May	710 kWh	690 kWh	High saving
June	860 kWh	760 kWh	High saving

Page 4 - Site Evidence

Review the site evidence below before answering questions about roof exposure and backup readiness.



Roof sunlight map

Backup Battery Snapshot		
Reserve target: 40%		
Estimated backup time: 9 hours		
Backup load	Energy	Priority
Refrigerator	0.8 kWh	High
Wi-Fi router	0.1 kWh	High
Air conditioner	3.2 kWh	Low

Battery dashboard